Year 10
Pathway 2
Test 6
2018

Baldivis
Secondary College

Simple & Compound Interest

Name: Solutions Class: ______ Time: 30 minutes Total Mark:____/28

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

Simple Interest

Question 1	**4 marks (2, 2)**

Calculate the **amount of Simple interest** and the **total repayment** on the following.

a) $1000 is borrowed for 2 years and the interest rate is 5% pa.

$I = pRT$

$p = 1000$
$r = 0.05$
$t = 2$

$= 1000 \times 0.05 \times 2$
$= \$100$ ✓

$A = 1000 + 100 = \$1100$ ✓

b) A rate of 7.5% pa over a year on $4700.

$I = prt$

$P = 4700$
$r = 0.075$
$t = 1$

$= 4700 \times 0.075 \times 1$
$= \$352.50$ ✓

$A = 4700 + 352.50 = \$5052.50$ ✓

Question 2	**2 marks**

$5500 is borrowed at 9.5% pa simple interest. $2090 in interest is repaid with the loan. How long did it take to repay the loan?

$p = 5500$
$r = 0.095$
$I = 2090$
$t =$

$I = prt$
$2090 = 5500 \times 0.095 \times t$ ✓
$t = 4$ years ✓

Question 3	2 marks

At what simple interest rate will an investment of \$6500 grow to a total investment of \$9750 in 8 years?

Interest = 9750 - 6500 = 3250

$r = ?$
$t = 8$ yrs
$p = 6500$

$I = prt$

$3250 = 6500 \times r \times 8$

$r = 0.0625$

6.25%

Compound Interest

Question 4	3 marks

How much will \$8000 be worth if it has been invested for 3 years with an interest rate of 2% p.a. compounding yearly? Give your answer correct to the nearest dollar.

$P = 8000$
$t = 3$
$r = 0.02$
$n = 1$

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$= 8000\left(1 + \frac{0.02}{1}\right)^{1 \times 3}$

$= \$8489.66$

$= \$8489$ $\frac{1}{2}$

Question 5	6 marks (3, 3)

A \$10 000 investment is placed in a bank returning 6% compound interest. Find the value of the investment, correct to the nearest cent, after a year if interest is calculated every:

(a) six months

$P = 10000$
$r = 0.06$
$t = 1$
$n = 2$

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$= 10000\left(1 + \frac{0.06}{2}\right)^{2 \times 1}$

$= \$10,609.00$

(b) month.

$n = 12$

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$= 10000\left(1 + \frac{0.06}{12}\right)^{12 \times 1}$

$= \$10616.78$

Question 6	6 marks (3, 3)

$25000 is invested for 4 years. Calculate the compound interest earned on each of the following.

$P = 25000$
$t = 4$

$A = P\left(1+\frac{r}{n}\right)^{nt}$

a) 7% pa compounded annually.

$n = 1$
$r = 0.07$

$= 25000\left(1+\frac{0.07}{1}\right)^{1\times 4}$
$= \$32769.90$

$I = A - P$
$= 32769.90 - 25000$
$= \$7769.90$

$A = P\left(1+\frac{r}{n}\right)^{nt}$

b) 2.75% compounded monthly.

$n = 12$
$r = 0.0275$

$A = 25000\left(1+\frac{0.0275}{12}\right)^{12\times 4}$
$= \$27903.44$

$I = A - P$
$= 27903.44 - 25000 = \$2903.44$

Compare Simple & Compound Interest

Question 7	5 marks (2, 2, 1)

Todd deposited $1,600 in an account that earns 7% simple interest every year.
His brother Adam deposited $1,550 in an account that earns 7.5% interest compounded annually.

The deposits were made on the same day, and no additional money will be deposited or withdrawn from the accounts.

Who will have the most money in their account and how much more at the end of 4 years?

TODD

SI

$I = prt$ $P = 1600$

$P = 1600$
$r = 0.07$
$t = 4$

$I = 1600 \times 0.07 \times 4$
$= \$448$

$A = 1600 + 448$
$= \$2048$

ADAM

$A = P\left(1+\frac{r}{n}\right)^{nt}$.

$P = \$1,550$
$r = 0.075$
$t = 4$ years
$n = 1$

$A = 1550\left(1+\frac{0.075}{1}\right)^{1\times 4}$
$= \$2069.98$

Adam will have the most money in his account $\frac{1}{2}$

$2069.98 - 2048 = \$21.98$ $\frac{1}{2}$

End of Test